THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MSBA7040 Sports Analytics

GENERAL INFORMATION

Instructor: Minje PARK

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Office: KK 831
Consultation times: TBD

Tutor: TBD

Pre-requisites: N/A

Course Website: Moodle

* This course does not require programming knowledge or mathematical skills.

COURSE DESCRIPTION

Today's sports industry has grown tremendously, raising total annual revenue of US$2.6 trillion worldwide. Professional sports teams invest hugely. For example, the LA Dodgers signed a US$700 million 10-year contract with Shohei Ohtani, the Golden State Warriors paid US$208 million to their players in the 2023-24 season, and Real Madrid spent US$1.9 billion to renovate its stadium, Santiago Bernabeu. To maximize the return on investment, today's sports teams leverage data and analytics to improve athletic performance and strategic decision-making.

This course introduces how data are used and analyzed in modern sports, such as basketball, baseball, football, and table tennis. It is designed to help master students without any background in analytics, mathematics, or statistics learn and understand the fundamentals of business analytics, using sports as an application area. Lectures will cover basic concepts of probability and different analytical tools, including regression, simulation, game theory, and machine learning. Although this course will not use any programming language, students will do several hands-on exercises and assignments handling data using Excel spreadsheets.

COURSE OBJECTIVES

1. Understand and describe various data from different sports
2. Apply statistical and analytical methods to sports data
3. Analyse sports data for improving player and team performance
4. Generalise sports analytics concepts and methods to other business problems

PROGRAMME LEARNING OUTCOME

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: Develop understanding of sports data and its use in the sports industry across the world	PLO 1, 4
CLO2: Identify and formulate the relevant sports analytics problem	PLO 1, 2
CLO3: Select and use effective methods to address the sports analytics problem	PLO 2, 3
CLO4: Communicate analytical issues and solutions to others	PLO 3, 5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected Hour	Study Load (% of study)
T&L1. Interactive lectures	30	25%
T&L2. Assignments	30	25%
T&L3. Course Project	30	25%
T&L4. Self-study	30	25%
Total	120	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Participation & Attendance		20%	CLO1, 2, 3, 4
A2. Assignments		50%	CLO1, 2, 3, 4
A3. Group Project		30%	CLO1, 2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT	
Course Grade Descriptors	
A+, A, A-	•Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •High participation in discussions •Present arguments that have an element of originality •Achieve a standard of excellent performance in assignments and course projects with very accurate computation and very good analytical and problem-solving skills •Excellent performance in assignments
B+, B, B-	•Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Good participation in discussions •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in assignments and course projects with accurate computation and good analytical and problem-solving skills •Good performance in assignments
C+, C, C-	•Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Some participation in discussions •Present arguments in a well-structure manner •Meet a standard of acceptable performance in in assignments and course projects with reasonably accurate computation and acceptable analytical and problem-solving skills •Acceptable performance in assignments
D+, D	•Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Minimal or no participation in discussions •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in assignments and course projects with some errors in computation and barely adequate analytical and problem-solving skills •Marginally acceptable performance in assignments
F	•Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Minimal or no participation in discussions •Present arguments poorly •Fail to meet a standard of passing assignments and course projects with major errors in computation and inadequate analytical and problem-solving skills •Poorly performance in assignments
Assessment Rubrics for Participation	
A+, A, A-	•High participation in discussions •Always attend in-class discussions •Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •Present arguments that have an element of originality •Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	•Good participation in discussions •Often attend in-class discussions •Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Present arguments that go beyond the lecture and textbook •Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	•Some participation in discussions •Sometimes attend in-class discussions •Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Present arguments in a well-structure manner •Respect others and follow the class rules (no chatting and do not use cell phone)

D+, D	•Minimal or no participation in discussions •Rarely attend in-class discussions •Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Present arguments in a marginally acceptable manner •Respect others and follow the class rules (no chatting and do not use cell phone)
F	•Minimal or no participation in discussions •Almost never attend in-class discussions •Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Present arguments poorly •Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments	
A+, A, A-	•Demonstrate a strong understanding of all relevant knowledge •Present arguments that have an element of originality •Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	•Demonstrate a good understanding of all relevant knowledge •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	•Demonstrate a basic understanding of the concepts involved •Present arguments in a well-structure manner •Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	•Demonstrate a minimum understanding of the concepts involved •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	•Demonstrate a poor understanding of the concepts involved •Present arguments poorly •Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

Session	Topic	Readings (optional)
1	***Introduction*** • Why sports analytics? • Understanding the rules and data of basketball, football, and baseball	
2	***Basketball Analytics 1*** • Shooting performance: probability • NBA's 3-point revolution: expectation • +/- rating	• Book: Ch. 30 • Video: How the 3-point line is breaking basketball
3	***Baseball Analytics 1*** • Prediction of winning • Evaluating pitchers: correlation • Evaluating hitters: regression	• Book: Ch. 1, Ch. 2, Ch. 5
4	***Basketball Analytics 2*** • Testing Stephen Curry's hot hand: Normal distribution • Hypothesis testing • Testing for NBA Referee bias	• Book: Ch. 11, Ch. 35
5	***Baseball Analytics 2*** • Evaluating players: simulation • Player salaries	• Book: Ch 4, Ch. 9
6	***Football Analytics*** • Penalty kicks and game theory • Norm theory for goalies' behavior • Pep Guardiola's zones: Markov chains	• Book: Ch. 39 • Video: How Pep Sees Football Differently
7	***Machine Learning with Fan Data*** • Basics of machine learning • Prediction models for ticket sales	• Case (required): Data Science at the Warriors
8	***Strategies in Sports*** • Analysis of bunting, stolen bases • When to foul • Field goal or first down	• Book: Ch. 6, Ch. 21, Ch. 38
9	***Other Sports and AI*** • Table tennis • Golf: Every shot counts • AI in sports	
10	***Course Wrap-up***	
11	***Group project competition***	

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

Textbook (optional):

Mathletics: How gamblers, managers, and fans use mathematics in sports. By Winston, W. L., Nestler, S., & Pelechrinis, K. (2022). Princeton University Press.

Case (required):

Data science at the Warriors. By Bojinov, I. I.,& Parzen M. I., Harvard Business School Case 622-048, August 2021.

Handouts and assignments will be posted on Moodle.

Article/Book

Bar-Eli, M., Azar, O. H., Ritov, I., Keidar-Levin, Y., & Schein, G. (2007). Action bias among elite soccer goalkeepers: The case of penalty kicks. *Journal of economic psychology,* 28(5), 606-621.

Broadie, M. (2014). Every shot counts: Using the revolutionary strokes gained approach to improve your golf performance and strategy. Avery.

Calmet, G., Erades, A., & Vuillemot, R. (2023). Exploring Table Tennis Analytics: Domination, Expected Score and Shot Diversity. In International Workshop on Machine Learning and Data Mining for Sports Analytics (pp. 167-178). Cham: Springer Nature Switzerland.

Chiappori, P. A., Levitt, S., & Groseclose, T. (2002). Testing mixed-strategy equilibria when players are heterogeneous: The case of penalty kicks in soccer. *American Economic Review*, 92(4), 1138-1151.

Gilovich, T., Vallone, R., & Tversky, A. (1985). The hot hand in basketball: On the misperception of random sequences. *Cognitive psychology*, *17*(3), 295-314.

Price, J., & Wolfers, J. (2010). Racial discrimination among NBA referees. *The Quarterly Journal of Economics*, 125(4), 1859-1887.

YouTube Video

How Pep Sees Football Differently | What Separates Pep Guardiola | Pep Guardiola's 20 Zone System. https://www.youtube.com/watch?v=jVE3Sqatdlg&t=280s

How the 3-point line is breaking basketball. https://www.youtube.com/watch?v=2p3NIR8LYoo.

Momentum isn't Magic – Hot Hand Analysis. https://www.youtube.com/watch?v=CR5vT44ZMK8.

The Math Behind Basketball's Wildest Moves | Rajiv Maheswaran | TED Talks. https://www.youtube.com/watch?v=66ko_cWSHBU.

The NBA Data Scientist. https://www.youtube.com/watch?v=MpLHMKTolVw.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

√ conducting mid-term survey in additional to SFTL around the end of the semester

- Online response via Moodle site
- Others: _____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

The University Regulations on academic dishonesty will be strictly enforced. Please check the University Statement on plagiarism at http://www.hku.hk/plagiarism/.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)